BANK OF UGANDA | EXECUTIVE MASTERCLASS

5-Day Cyber Forensics & Security Investigations

- **Institutional Mandate:** Guarding Uganda's sovereign payment rails, RTGS, and interbank settlement interfaces.
- **Complete Forensic Lifecycle:** Evidence seizure, RAM volatile extraction, C2 telemetry, containment & judicial reporting.
- **Statutory Defensibility:** Built upon Uganda Computer Misuse Act (2011/2022) & Electronic Signatures Act 2011.
- **Hands-on Case Study:** 'Operation Crested Crane' 4.8 Billion UGX settlement breach drill.

DAY 1: FOUNDATIONS & EVIDENTIARY INTEGRITY

Principles of Forensic Soundness & Order of Volatility

- **Order of Volatility (RFC 3227):** CPU Registers -> RAM -> Swap -> NVMe/SAN -> Backup Tapes.
- **ISO/IEC 27037 Compliance:** Never analyze original evidence; duplicate bitstreams under hardware write-blockers.
- **Cryptographic Redundancy:** Simultaneous dual hashing (SHA-256 + MD5) to guarantee immutability.
- **Forensic Trap Alert:** Binary mode ('wb' / 'rb') capture prevents OS CRLF conversion from corrupting legal hashes.

DAY 2: NETWORK FORENSICS & TELEMETRY

Financial Cyber Attack Vectors & Beacon Dissection

- **C2 Beacon Statistical Triage:** Low jitter (< 0.15) flags automated beacon heartbeats vs. organic human browsing.
- **Payment Gateway Probes:** Detecting unauthorized outbound wire payloads and rogue beneficiary IDs.
- **Protocol Telemetry:** Full packet dissection (Wireshark), Zeek metadata logs, and TLS JA3/JA4 fingerprinting.
- **Lateral Movement Detection:** Kerberoasting, Pass-the-Hash, and PsExec activity in financial subnets.

DAY 3: HOST ARTIFACTS & MEMORY FORENSICS

Forensic Imaging & Volatile Memory Triage

- **Volatile RAM Forensics (Volatility 3):** Inspecting unlinked stealth processes ('psscan') and injected DLLs ('malfind').
- **Windows Endpoint Forensics:** $MFT timestamps, Prefetch execution traces, Shimcache, and Event ID 4624/4688 logs.
- **Anti-Forensics Unmasking:** Identifying timestomping ($STANDARD_INFO vs $FILE_NAME discrepancies).
- **Data Carving & Extraction:** Reconstructing deleted transactions and temporary staging buffers.

DAY 4: INCIDENT RESPONSE & FINANCIAL MALWARE

BoU CSIRT Playbooks & Reverse Engineering Triage

- **Containment Without Reboot:** Isolating infected hosts on VLANs to safeguard volatile in-memory evidence.
- **Active Directory Eviction:** Golden Ticket eradication via dual 'krbtgt' password rotation within 10 hours.
- **Banking Trojan Anatomy:** Web-injects, credential scraping, and in-memory payment API hooking.
- **Threat Signature Engineering:** Writing custom YARA rules targeting financial malware strings and headers.

DAY 5: UGANDA CYBER LAWS & REPORTING

Statutory Admissibility & Executive Briefings

- **Uganda Computer Misuse Act (2011/2022):** Section 12 (Unauthorized Access), Section 15 (Modification), Section 20 (Fraud).
- **Data Protection and Privacy Act 2019:** Ensuring custody handling safeguards non-incident financial data.
- **Courtroom Testimony Standards:** Demonstrating unbroken chain of custody and scientifically validated toolsets.
- **Executive Dossier Delivery:** Briefing the Governor, Board Audit Committee, and Judicial Prosecutors.